

Practice Final Exam 1

Name
Date
Arithmetic

Instructions

- I) Write a title, name, date, and course.
- II) Number each problem, circle the number, rewrite the problem, and show all work.
- III) Put a rectangle around your answers.
- IV) Write any fractions in your answers in lowest terms and never leave your answers as improper fractions.

① Write the following number in word form.

a) 2,498,762

Write $>$, $<$, or $=$ between the pairs of numbers below.

b) 7,800 7,799 c) 1.9000 1.9 d) 0.5999 0.6883

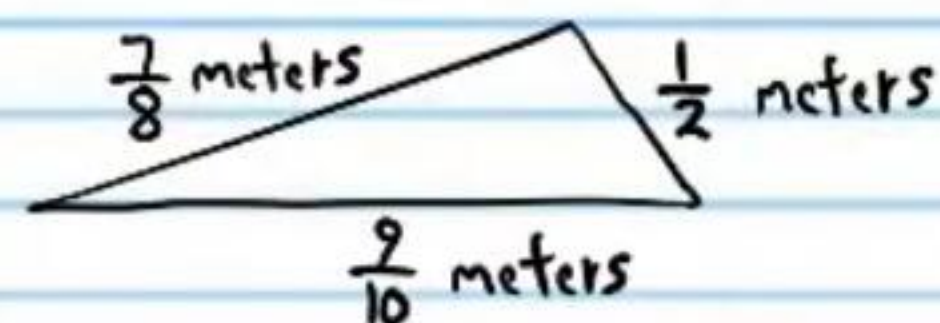
② i) The quotient of a number and six decreased by one is eight. Which of the following could be the number?

- A 54
- B 48
- C 42
- D 36
- E 30

ii) A student scored 6, 4, and 7 on his first three quizzes. What will he have to score on his next quiz to raise his average score to 6?

- A 9
- B $5\frac{2}{3}$
- C 1
- D 7
- E 8.5

③ Find the perimeter of the triangle below.



* Sides not drawn to scale.

④ a) Find the prime factorization of 45.

b) Find the first four multiples of seven.

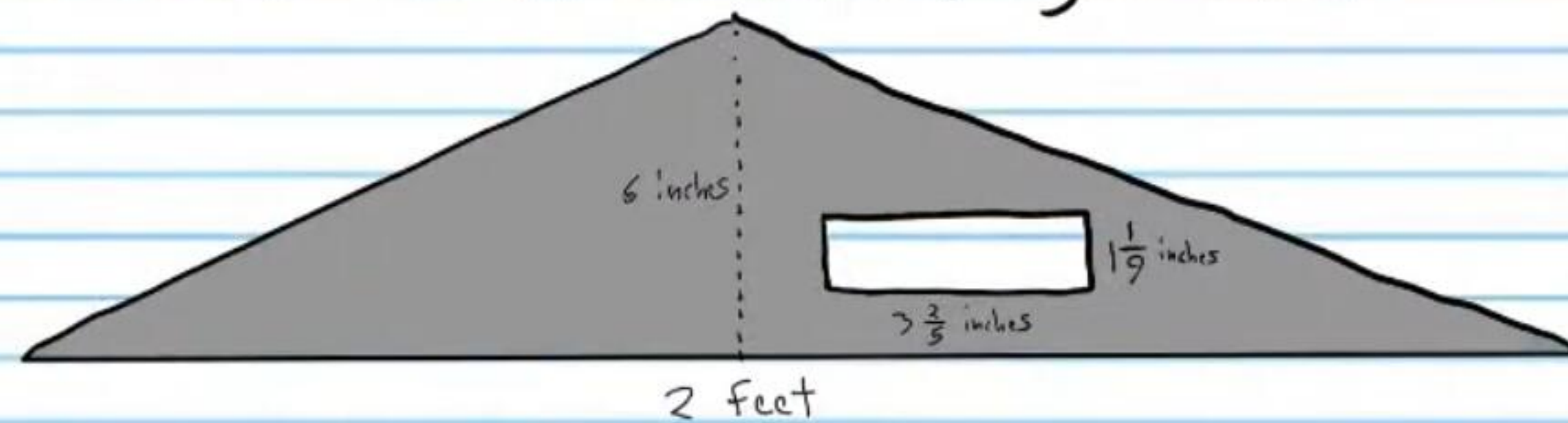
⑤ a) What is the GCF of 63 and 42?

b) What is the LCM of 9 and 6?

⑥ Jerry had $5\frac{3}{4}$ ounces of silver. He sold $4\frac{3}{4}$ of the silver. How much does he have now?

⑦ A rabbit hops $6\frac{3}{4}$ miles per hour. If this rabbit hopped $1\frac{4}{5}$ miles without stopping, for what fraction of an hour was he hopping?

⑧ Find the area of the shaded region below.



⑨ Jill makes \$16.20 per hour at work. If she works for 31.4 hours, how much money will she make?

⑩ You share a bag of popcorn with two friends. Initially, each person gets an equal share, but Alice gives $\frac{3}{8}$ of her share to Brad, and you give $\frac{1}{2}$ of your share to Brad. What fraction of the original bag of popcorn did Brad get (reference class 35, #7A-D)?

⑪ Jesse had $3\frac{5}{6}$ pounds of nails, 8.7 pounds of tools, $\frac{2}{3}$ of a pound of concrete, and 10 pounds of tiles. How much do all these items weigh together?

⑫ a) Convert the following fraction to a decimal, then to a percent: $\frac{1}{8}$.

b) Convert the following percent to a decimal, then to a fraction: 0.013 %.

⑬ Tina found 1.8007 grams of copper, 23 grams of iron, and 42.3 grams of calcium carbonate. What is the total weight of her findings?

⑭ What is 83% of 36%?

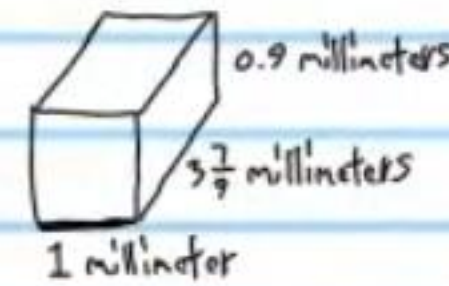
⑮ Approximate the volume of a sphere with a radius of four feet using the formula below and round your answer to the nearest thousandth.

$$V \approx 4 \times 3.14 \times r^3 \div 3$$

⑯ An object moves 20.05 centimeters in 0.008 hours. Find its unit rate. Round your answer to the nearest whole number.

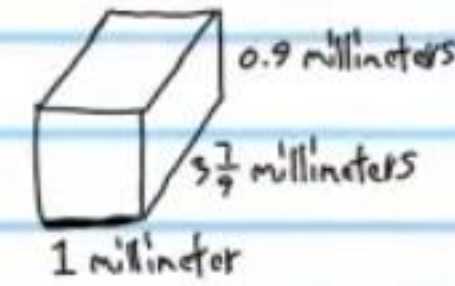
In the following three multiple choice problems, choose the fastest way to find the answers. Write "E" for estimation, "AS" for algorithmic shortcut, or "A" for algorithm.

⑰A Use the following formula to find the volume of the rectangular prism: $V = L \times W \times H$.



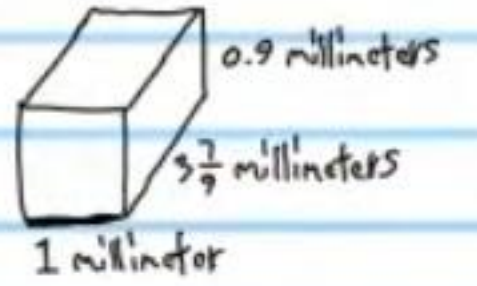
- A $3\frac{3}{10} \text{ mm}^3$
- B $3\frac{4}{10} \text{ mm}^3$
- C $3\frac{5}{10} \text{ mm}^3$
- D $3\frac{6}{10} \text{ mm}^3$
- E $3\frac{7}{10} \text{ mm}^3$

⑰B Use the following formula to find the volume of the rectangular prism: $V = L \times W \times H$.



- A 0.0009 mm^3
- B 3.4 mm^3
- C 67 mm^3
- D 1 mm^3
- E 5.7 mm^3

⑰C Use the following formula to find the volume of the rectangular prism: $V = L \times W \times H$.

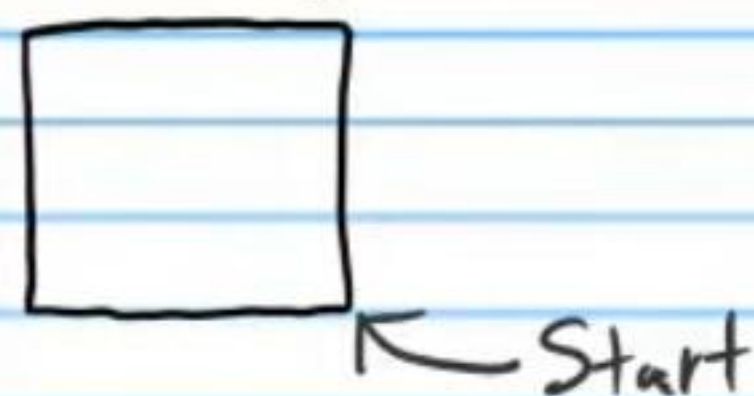


- A $3\frac{1}{5} \text{ mm}^3$
- B $3\frac{2}{5} \text{ mm}^3$
- C $3\frac{3}{5} \text{ mm}^3$
- D $3\frac{4}{5} \text{ mm}^3$
- E 4 mm^3

Masses of Common Coins (grams): 5, 2.5, 3, 4.

⑰ Find the mean, median, mode, range, and interquartile range of the data on coin masses above. Draw a dot plot of the data and a box and whisker plot above it. Identify any outliers.

- ①9 An object travels on the sides of the shape below. It takes the object three seconds to move from one corner to the next. If the object starts on the bottom right corner, and moves clockwise, at which side will it be after traveling for 407 seconds?



- ②0 Write $<$, $>$, or $=$ between the pairs of fractions below.

a) $\frac{2}{7} \square \frac{5}{14}$ b) $\frac{3}{3} \square \frac{8}{12}$ c) $\frac{3}{5} \square \frac{5}{8}$ d) $\frac{49}{56} \square \frac{64}{72}$

- ②1 You deposit \$36 in a checking account that pays 0.8 percent simple interest annually. What will your account balance be after four years, assuming that you make no additional deposits? Round your answer to the nearest hundredth (i.e. to the nearest cent).

- ②2 Alan had a growth spurt during the summer. His height increased by 6%. If his height before the growth spurt was 52 inches, what is his new height?

- ②3 Arrange the following numbers from least to greatest.

$3\frac{2}{5}$, 3.4, 3.25, $3\frac{3}{10}$

- ②4 The ratio of red hats to blue hats is 6 to 5. If there are 35 blue hats, how many red hats are there? Write a proportion statement to represent this problem and use it to find the answer.

- ②5 Write the rules/properties demonstrated by the following equations

a) $(2+3)+5 = 2+(3+5)$ b) $0 = 0 \times 100$ c) $1 = 3 \times \frac{1}{3}$ d) $3 \times 5 = 5 \times 3$